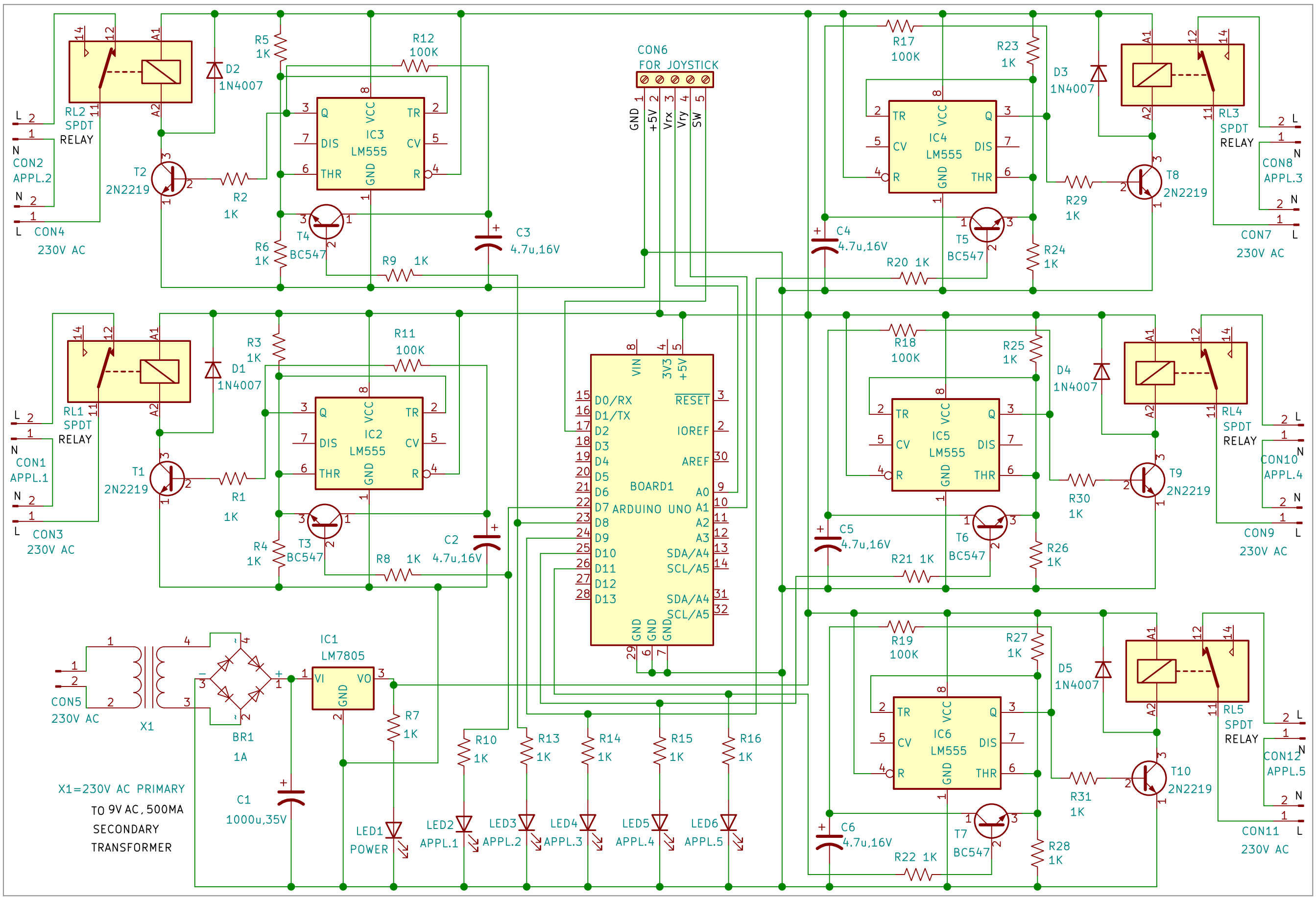


Fig. 2: Circuit diagram

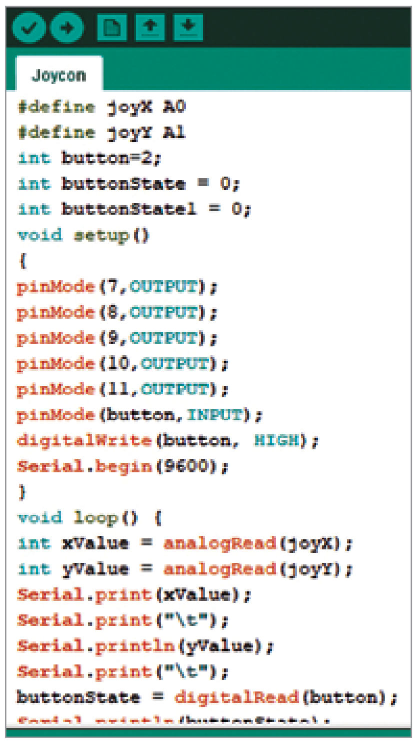
LED1 indicates that power supply is available and the circuit is enabled. Joysticks come in various shapes and sizes, and the one used here is shown in Fig. 3.

A joystick module typically provides analogue outputs, which change as the joystick is moved in different directions. Its movement can be interpreted in two axes: X-axis and Y-axis. Movement in each axis is represented by a potentiometer, and their midpoints are driven out as Rx and Ry. These potentiometers act as voltage dividers when the joystick is in standby. When the joystick is moved, the voltage on each pin changes depending on the direction, representing four directions of joystick movement based on two ADC outputs.

The joystick module used here also has two potentiometers inside it: one for X-axis movement and another for Y-axis movement. Each potentiometer receives 5V from the Arduino. As the joystick moves, the voltage value changes, and the analogue value at pins A0 and A1 also changes. From the Arduino, we read the analogue values for the X and Y axes and turn on the LEDs accordingly. A push-button switch on the joystick module controls LED2 in the circuit.

Fig. 3: Joystick

There are five toggle circuits built around LM555 (IC2 through IC6), all identical. Let us take a look at IC2, which toggles relay RL1 when you press the joystick. Pins 2 and 6, the threshold and trigger inputs, are held at half the supply voltage by the two



Joycon

```
#define joyX A0
#define joyY A1
int button=2;
int buttonState = 0;
int buttonState1 = 0;
void setup()
{
pinMode(7,OUTPUT);
pinMode(8,OUTPUT);
pinMode(9,OUTPUT);
pinMode(10,OUTPUT);
pinMode(11,OUTPUT);
pinMode(button,INPUT);
digitalWrite(button, HIGH);
Serial.begin(9600);
}
void loop() {
int xValue = analogRead(joyX);
int yValue = analogRead(joyY);
Serial.print(xValue);
Serial.print("\t");
Serial.println(yValue);
Serial.print("\t");
buttonState = digitalRead(button);
```

Fig. 4: Source code